Homework 9

STAT 300 (Fall 2024)

Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirtname.nb.html> file by the deadline.

(1). Suppose you want to compare the average ages of cars among faculty,students and staff at Penn State. You take a random sample of 200 people who have parking permits and ask *how old is their primary car?* You ask your friends in STAT 300 about using ANOVA to analyze this data and below are some responses you received. Indicate how you would response to each of below. Your response should have explanation beyond simply yes/no, true/false.

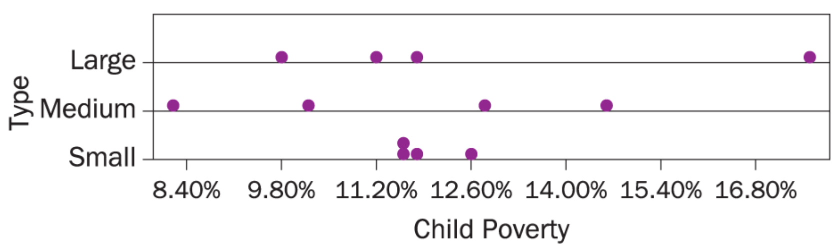
(a) You cannot use ANOVA as there are 3 groups.

(b) You cannot use ANOVA on this data because the response is quantitative.

(c) You cannot use ANOVA for this data because the sample size for the three groups will probably be different.

(d) You can do calculations for ANOVA on this data even though the sample sizes in groups will be different, but you cannot generalize the findings of the results to the population of people (staff, faculty and students) with parking permits at Penn State.

(2). Random sample of four counties taken from small, medium large counties in Iowa (total of 12 counties taken at random). We are interested if poverty rate in Iowa is associated with the size of the county. Below shown is a dotplot of the data. Does this plot raise any concerns with respect to use of ANOVA. Explain based on what you see in the dotplot.



Cannon, et. al., *STAT2: Modeling with Regression and ANOVA,* 2e, © 2019 W. H. Freeman and Company

(3). The Energy Information Administration gathers data on residential energy consumption. Random samples of households were taken from four regions in US. Data is provided in *energy.csv* file.

(a) Make appropriate graph(s) to see the relationship between the energy consumption and the region. Comment on your observations.

(b) Write down the one-way ANOVA model that you think will explain the relationship between the energy consumption and the region in the population. Define any terms you have used.

(c) Suppose we want to see whether or not there is a difference in average energy consumption among the four regions.

(i) Set up the appropriate null and alternative hypothesis.

(ii) Fit the model and generate the R output

(iii) Check the validity of the assumptions of the fitted model

(iv) Write the final conclusion in the context of the problem.

(d) Conduct any post-hoc analysis that might be relevant. When conducting the posthod analysis use a moderate approach (i.e.: not too conservative or not too liberal)

(e) Summarize your findings from the post-hoc analysis.